

Online Sales Data Report

- The dataset consists of 240 entries and 9 columns. There are no missing values, as all columns contain 240 non-null records. The data is stored in appropriate data types, including integer, float, and object types, ensuring data quality and consistency for analysis.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 240 entries, 0 to 239
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Transaction ID         240 non-null   int64  
 1   Date                  240 non-null   object  
 2   Product Category      240 non-null   object  
 3   Product Name          240 non-null   object  
 4   Units Sold            240 non-null   int64  
 5   Unit Price            240 non-null   float64 
 6   Total Revenue         240 non-null   float64 
 7   Region                240 non-null   object  
 8   Payment Method        240 non-null   object  
dtypes: float64(2), int64(2), object(5)
memory usage: 17.0+ KB
```

- The dataset includes 240 unique transactions recorded across 3 regions and 3 payment methods. There are 6 product categories and 232 unique products. The large number of unique product names suggests a diverse product portfolio, while the presence of only 3 regions and payment methods indicates a focused market and payment structure.

0	
Transaction ID	240
Date	240
Product Category	6
Product Name	232
Units Sold	7
Unit Price	117
Total Revenue	173
Region	3
Payment Method	3
dtype: int64	

- Transaction ID and Date contain no duplicate values, indicating that each transaction is uniquely identified and recorded on a unique date. Other columns contain duplicate values because multiple transactions may belong to the same product category, region, or payment method.

	Column	Duplicate_Count
0	Transaction ID	0
1	Date	0
2	Product Category	234
3	Product Name	8
4	Units Sold	233
5	Unit Price	123
6	Total Revenue	67
7	Region	237
8	Payment Method	237

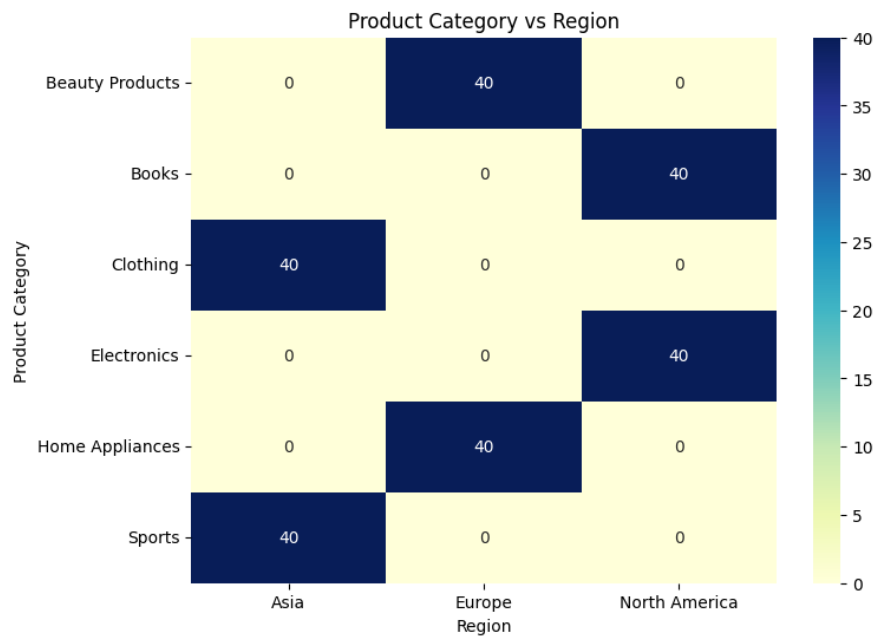
- The dataset contains 240 transactions. On average, 2.16 units are sold per transaction, with an average unit price of 236.40 and an average revenue of 335.70 per transaction. The minimum unit price recorded is 6.50, while the maximum unit price reaches 3899.99, indicating a wide range of products from low-cost to premium items.

	Transaction ID	Date	Units Sold	Unit Price	Total Revenue
count	240.00000	240	240.000000	240.000000	240.000000
mean	10120.50000	2024-04-29 12:00:00	2.158333	236.395583	335.699375
min	10001.00000	2024-01-01 00:00:00	1.000000	6.500000	6.500000
25%	10060.75000	2024-02-29 18:00:00	1.000000	29.500000	62.965000
50%	10120.50000	2024-04-29 12:00:00	2.000000	89.990000	179.970000
75%	10180.25000	2024-06-28 06:00:00	3.000000	249.990000	399.225000
max	10240.00000	2024-08-27 00:00:00	10.000000	3899.990000	3899.990000
std	69.42622	NaN	1.322454	429.446695	485.804469

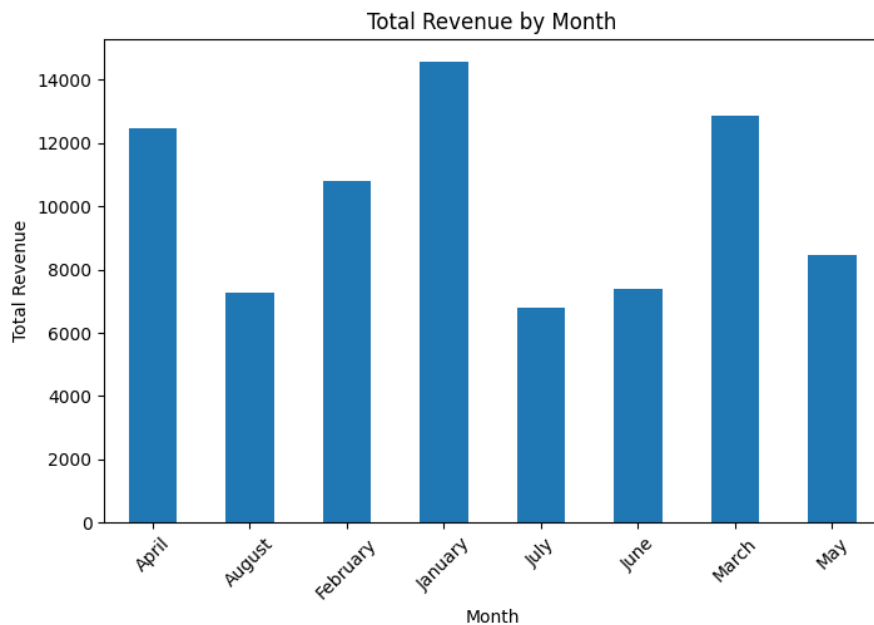
- The median number of units sold is 2, meaning that half of the transactions involve selling two or fewer units. Similarly, the median unit price is 89.99 and the median revenue is 179.97. The standard deviation of unit price (429.45) and total revenue (485.80) suggests significant variation in product prices and transaction values across the dataset.
- Furthermore, 75% of transactions have a unit price below 249.99 and generate revenue below 399.23, while the remaining 25% consist of higher-value transactions. The maximum revenue recorded for a single transaction is 3899.99.

	Transaction ID	Date	Units Sold	Unit Price	Total Revenue
count	240.00000	240	240.000000	240.000000	240.000000
mean	10120.50000	2024-04-29 12:00:00	2.158333	236.395583	335.699375
min	10001.00000	2024-01-01 00:00:00	1.000000	6.500000	6.500000
25%	10060.75000	2024-02-29 18:00:00	1.000000	29.500000	62.965000
50%	10120.50000	2024-04-29 12:00:00	2.000000	89.990000	179.970000
75%	10180.25000	2024-06-28 06:00:00	3.000000	249.990000	399.225000
max	10240.00000	2024-08-27 00:00:00	10.000000	3899.990000	3899.990000
std	69.42622	NaN	1.322454	429.446695	485.804469

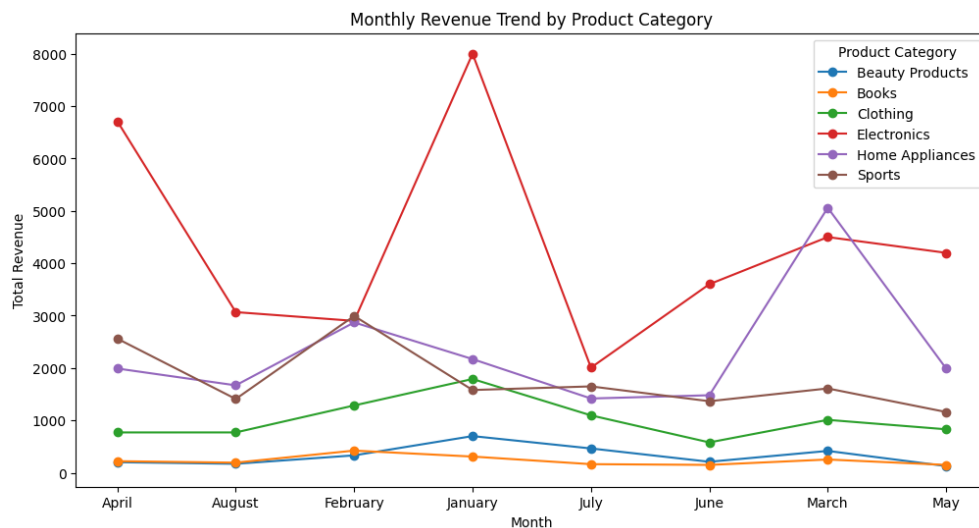
- Each product category has 40 transactions. Beauty Products and Home Appliances are linked to Europe, Books and Electronics to North America, and Clothing and Sports to Asia.



- January generated the highest revenue (14,548.32), while July recorded the lowest revenue (6,797.08). Overall, sales were strongest during the first four months of the year.



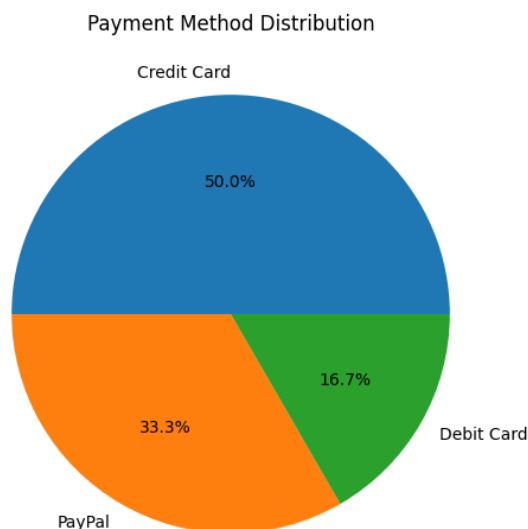
- Electronics was the highest revenue-generating category, reaching its peak in January and remaining one of the strongest performers throughout the period. Home Appliances also showed strong performance, with its highest revenue recorded in March. Clothing and Beauty Products achieved their maximum revenue in January, while Sports performed best in February. Books generated the lowest revenue among all product categories, despite reaching their highest revenue in February. Overall, the revenue trends indicate that different product categories experience peak sales in different months, suggesting seasonal variations in customer demand.



- Garmin Forerunner 945 generated the highest revenue (1,599.97) among the most frequently purchased products. Several products, including Keurig K-Elite Coffee Maker, Dyson Supersonic Hair Dryer, Sony WH-1000XM4 Headphones, and Dune by Frank Herbert, appeared in two transactions each. Most products in the dataset were purchased only once, indicating a diverse product portfolio with limited repeat purchases.

	Product Name	Total_Revenue	Count
79	Garmin Forerunner 945	1599.97	2
103	Keurig K-Elite Coffee Maker	529.97	2
56	Dune by Frank Herbert	71.96	2
58	Dyson Supersonic Hair Dryer	1199.97	2
172	Sony WH-1000XM4 Headphones	1049.97	2
181	The Catcher in the Rye by J.D. Salinger	88.93	2
185	The Girl with the Dragon Tattoo by Stieg Larsson	52.95	2
202	The Silent Patient by Alex Michaelides	134.95	2
8	Amazon Echo Dot (4th Gen)	199.96	1
9	Amazon Echo Show 10	249.99	1

- Credit Card was the most frequently used payment method (50%), followed by PayPal (33.3%) and Debit Card (16.7%).



- North America is the strongest-performing region, contributing nearly **46%** of total revenue. Asia and Europe contribute almost equally, with Asia having a small lead over Europe. This suggests that business performance is significantly stronger in North America compared to the other two regions.

Total Revenue	
Region	
Asia	22455.45
Europe	21268.06
North America	36844.34

dtype: float64

- The correlation analysis reveals a strong positive relationship between Unit Price and Total Revenue ($r = 0.93$), indicating that revenue is primarily influenced by pricing. Units Sold shows a weak negative correlation with both Unit Price ($r = -0.31$) and Total Revenue ($r = -0.17$), suggesting that sales volume has a relatively limited impact on overall revenue compared to pricing strategies.

